



DEVELOPER ARCHITECTURE FRAMEWORK

# Saathi

A digital dairy supply-chain & traceability platform for the Nand Baba  
Dugdh Mission • Uttar Pradesh / PCDF-Parag

CLASSIFICATION

**Strictly Confidential**

VERSION

**v0.1**

DATE

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PREPARED BY

**PYAAS**

**CONFIDENTIAL**

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# SAATHI — Developer Architecture Framework

## A digital dairy supply-chain & traceability platform for the Nand Baba Dugdh Mission (Uttar Pradesh / PCDF-Parag)

**Status:** v0.1 architecture blueprint for the engineering team. **Scope:** end-to-end system from a single cow to the consumer's QR scan, with same-day farmer payment, milk safety gating (aflatoxin/coliform), full batch traceability, cattle-health readiness, and government-grade compliance. **How to read this:** Sections 1–3 set context. Section 4 answers your *onboarding / one-phone-many-roles* question. Section 5 is the *role catalog* (which profiles to build, grounded in the real cooperative structure). Sections 6–13 are the architecture. Section 18 is the full *compliance register*. Section 20 lists decisions still needed.

### 1. Context (researched, not assumed)

Saathi is not being built in a vacuum. It must fit the **real** institutional and digital landscape:

- The mission.** Nand Baba Dugdh Mission is a ₹1,000-crore, 5-year UP Dairy Development Department programme (launched June 2023) to put a milk cooperative in every village, raise **per-animal yield**, promote indigenous breeds, and route subsidies via **DBT**. Saathi is the on-the-ground digitisation layer for this mission. Milk *traceability* and *disease-free milk* are stated goals — that is Saathi's core.
- The cooperative spine (Anand three-tier model, as run by PCDF).** Your hierarchy maps precisely onto the real structure:

| Your term                                           | Real cooperative entity                                                                         | Tier   |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| Samiti (with Adhyaksh + Sacheev)                    | <b>Dairy Cooperative Society (DCS)</b> — village level                                          | Tier 1 |
| Sangh ("Parambhiik Bahudeshiya Sahkari Samiti Ltd") | <b>Milk Union / Dugdh Utpadak Sahakari Sangh</b> — district level                               | Tier 2 |
| Federation / PCDF                                   | <b>Pradeshik Cooperative Dairy Federation (PCDF)</b> — state apex, the <b>Parag</b> brand owner | Tier 3 |

**⚠ Terminology note for the team:** "Parambhiik Bahuddeshiya Sahkari Samiti" is the name pattern used for **PACS/multipurpose primary societies**. In the *dairy* value chain the village body is the **DCS**. Model the org unit generically (see §5.1) so both labels map to the same node type — some villages run a multipurpose society that *also* collects milk.

- The existing national digital stack you must interoperate with (do NOT rebuild these):**
- Bharat Pashudhan / National Digital Livestock Mission (NDLM)** — the national animal database. Every animal gets a **12-digit "Pashu Aadhaar" ear-tag ID** (~35.7 crore issued as of Nov 2025) that is the *primary key* for vaccinations, AI, disease reports, and milk recording. **This is the hook for your cattle-collar "backdoor" (see §9).**
- 1962** — toll-free Mobile Veterinary Unit (MVU) helpline + the **1962 App** (~4,335 MVUs sanctioned). Saathi integrates, it does not replace.
- NDDB AMCS** (Automatic Milk Collection System) and **i-DIS** (Internet-based Dairy Information System) — existing DCS/Union software. Saathi should ingest/federate rather than duplicate where these are already deployed.

- **India Stack** — Aadhaar eKYC, DigiLocker, Account Aggregator, UPI/AePS, DBT-via-PFMS. Your KYC, bank-linking, and payment rails ride on these.

**Design consequence:** Saathi is best positioned as a **state-level (UP/PCDF) orchestration super-app** that federates the national DPI systems and adds the pieces they lack — real-time traceability, safety gating, same-day settlement, and a low-literacy multi-role UX.

## 2. Objectives → engineering targets

| Business goal                            | Engineering translation                                                                                                        |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Make milk traceable end-to-end           | Immutable, append-only <b>provenance graph</b> : pour → DCS lot → route → BMC → plant batch → product QR (§7)                  |
| Disease-free milk (aflatoxin, coliform)  | <b>Quality gates</b> at DCS/BMC/plant that can <i>block</i> a lot from advancing; FSSAI thresholds enforced in software (§8.3) |
| Same-day farmer payment                  | Autofilled invoice on the day of pour + settlement rail (§8.1); reduce today's 10-day cycle                                    |
| One government app, every phone = a role | <b>Party + org-scoped RBAC</b> identity model with a role-switcher (§4)                                                        |
| Illiterate-farmer-friendly               | Voice, icon-first, vernacular, offline-first UX (§16)                                                                          |
| Increase yield per cow                   | Cattle profile + health/AI records + analytics, collar-ready (§9)                                                              |
| Digitise records for government          | Read APIs + dashboards for Mission/PCDF/district officials, audit-grade logs (§12, §18)                                        |

## 3. Design principles & hard constraints (from the field)

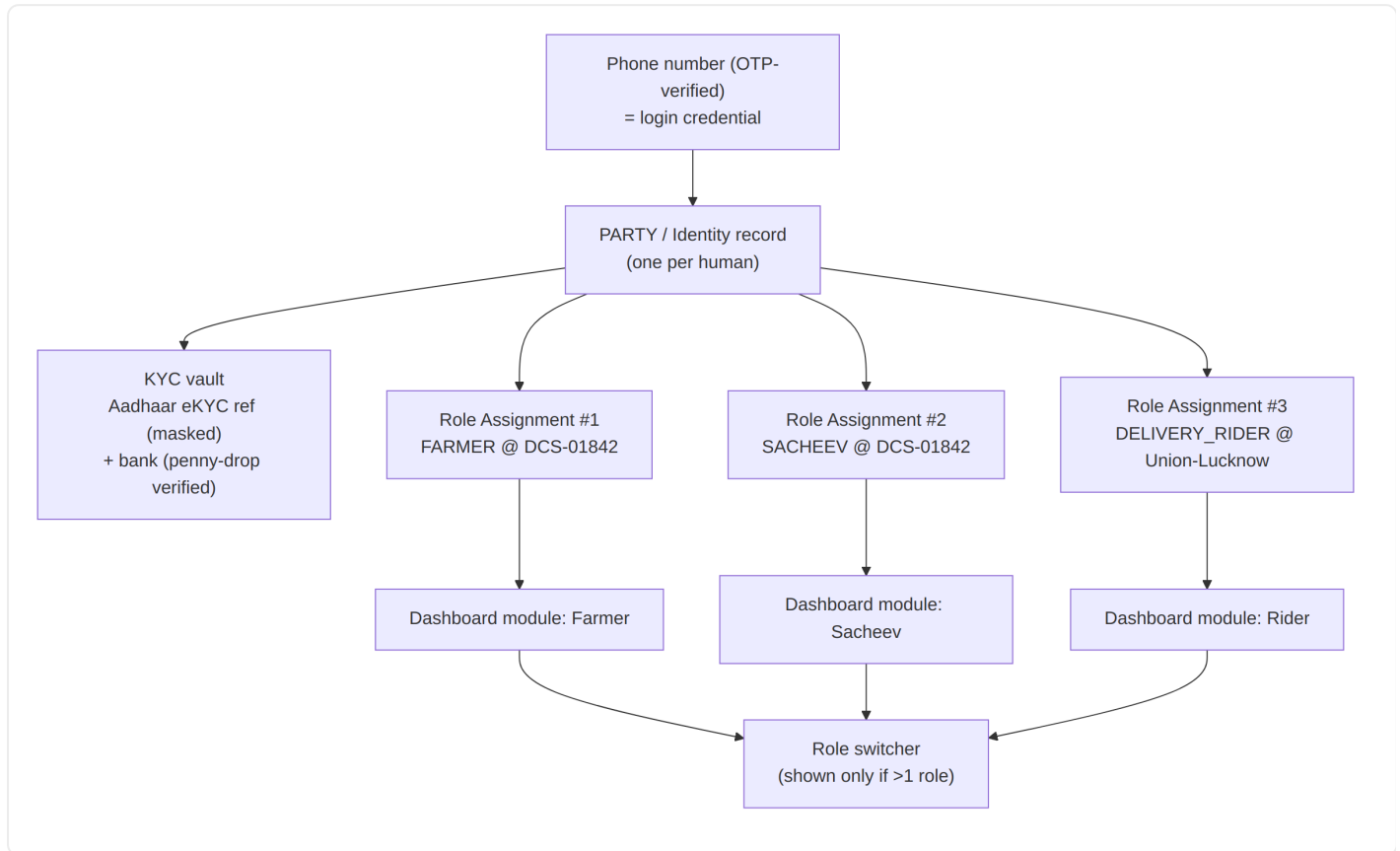
The NDDDB AMCU field spec is explicit about the environment. Build for it:

1. **Offline-first, sync-later.** Villages have unreliable power and connectivity. Every collection-point action must work fully offline and reconcile on reconnect. Treat the network as *optional*.
2. **Low/no digital literacy.** Assume the farmer cannot read fluently. Assume the Sacheev is *not* IT-savvy. Icon + voice + vernacular, not text forms.
3. **DPI-first, don't duplicate.** Reuse Aadhaar/DigiLocker/AA/UPI/Bharat Pashudhan. Every avoided rebuild is avoided liability.
4. **Provenance is append-only.** Traceability records are never edited in place — corrections are new events referencing the old (audit + tamper-evidence).
5. **Tamper-awareness.** *A photo of a screen is spoofable.* Wherever human-entered or photo-derived data enters, add integrity controls (§8.2).
6. **Capability gating over schema churn.** Features that depend on future schemes (collars) are provisioned dormant behind feature flags, not bolted on later.
7. **Pooling reality.** Once milk is poured into a common can and pooled at BMC/plant, an individual *pack* traces to a **set of contributing samitis**, not one farmer. Design honesty into the data model (§7.4).

## 4. Identity & onboarding — "one phone number, one identity, many roles"

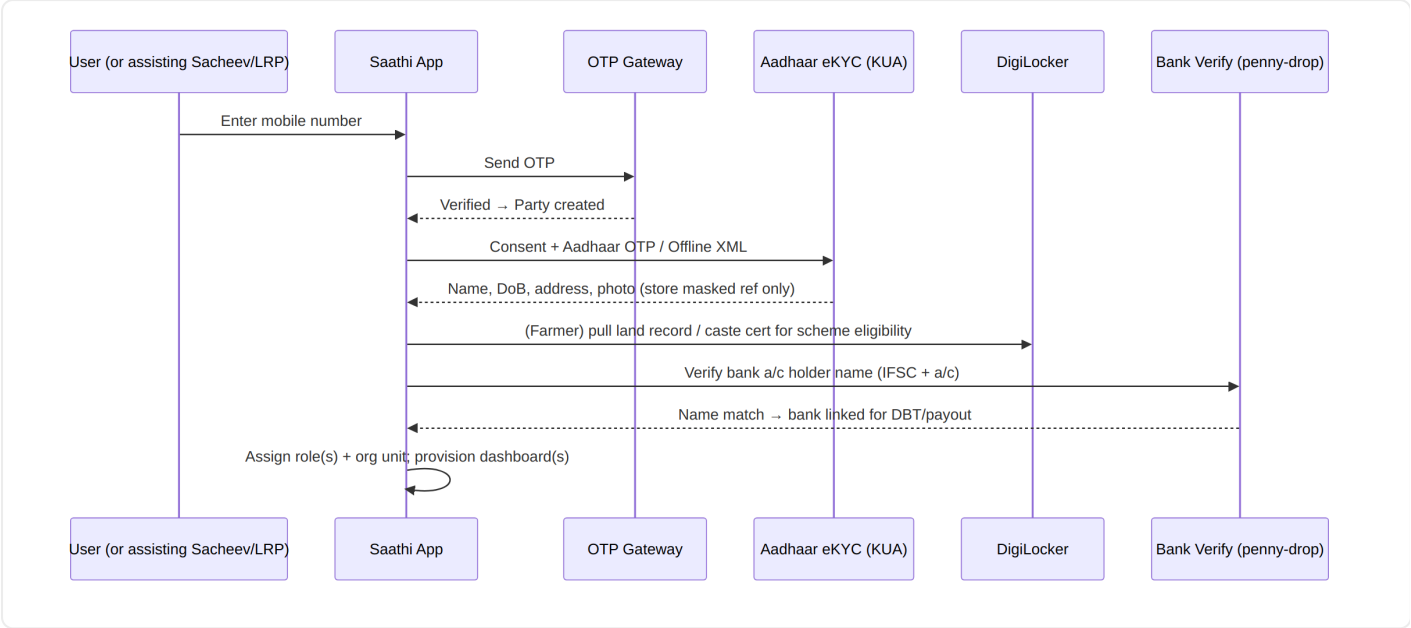
This is the crux of your onboarding question. The answer is a **Party model with org-scoped Role-Based Access Control**, not one account type per persona.

### 4.1 The model



**Why this shape:** - A Sacheev *is also* a farmer. An Adhyaksh *is also* a farmer. A rider might later become a plant operator. One human = one Party = many time-bounded, org-scoped **Role Assignments**. The app resolves all roles for the logged-in phone number and renders a **role switcher**; each role lazy-loads its own dashboard module inside the *single* app. - Every role assignment is scoped to an **org unit** (a specific DCS, Union, or the Federation) and carries a validity window, so revoking a Sacheev when they lose an election is a single record update — not an account deletion.

4.2 Onboarding flow (assisted, tiered KYC)



KYC assurance tiers:

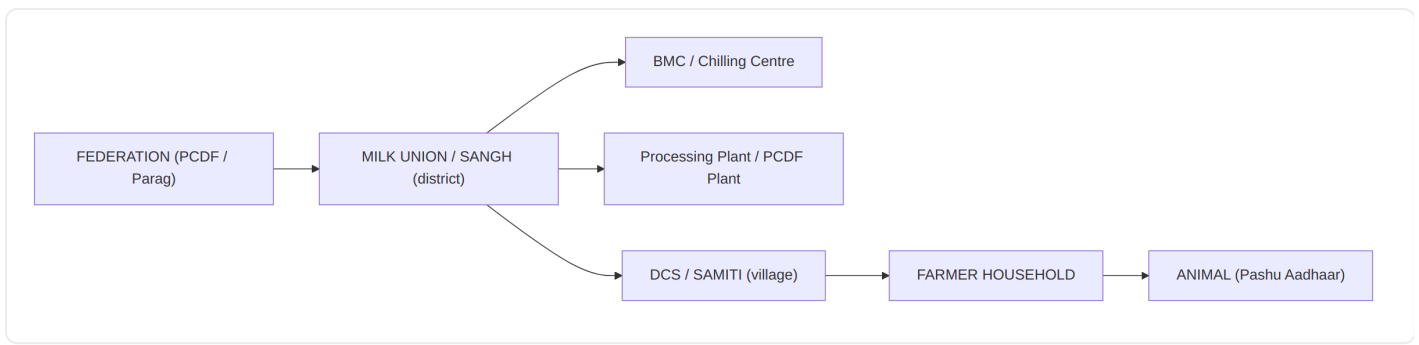
| Persona                      | Identity proof                                                                               | Bank                  | Extra                                                 |
|------------------------------|----------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------|
| Farmer / producer            | Aadhaar OTP eKYC <b>or</b> offline XML (never make Aadhaar <i>mandatory</i> to deny service) | Penny-drop name match | DigiLocker land/caste docs for scheme DBT eligibility |
| Sathive / Adhyaksh           | Aadhaar eKYC + cooperative membership record                                                 | Society a/c mapping   | Election record reference                             |
| Union/PCDF/Mis sion official | Department employee ID mapping + Video-KYC (V-CIP)                                           | Govt payroll a/c      | Role granted by admin, not self-serve                 |
| Rider (van / last-mile)      | Aadhaar eKYC + DL/RC for vehicle                                                             | Payout a/c            | Background/gig onboarding                             |
| Consumer                     | Mobile OTP only (minimal data)                                                               | —                     | No Aadhaar unless transacting                         |

**Compliance hooks baked in here:** Aadhaar first-8-digit masking + Aadhaar Data Vault for any stored reference; multilingual plain-language consent notice; consent logged and withdrawable; **never** block service for absence of Aadhaar (§18-A).

5. Role & persona catalog (which profiles to build)

Grounded in the *actual* cooperative + national-stack roles. Build these as RBAC roles; each maps to a dashboard module.

## 5.1 Organisational-unit model (the backbone RBAC is scoped to)



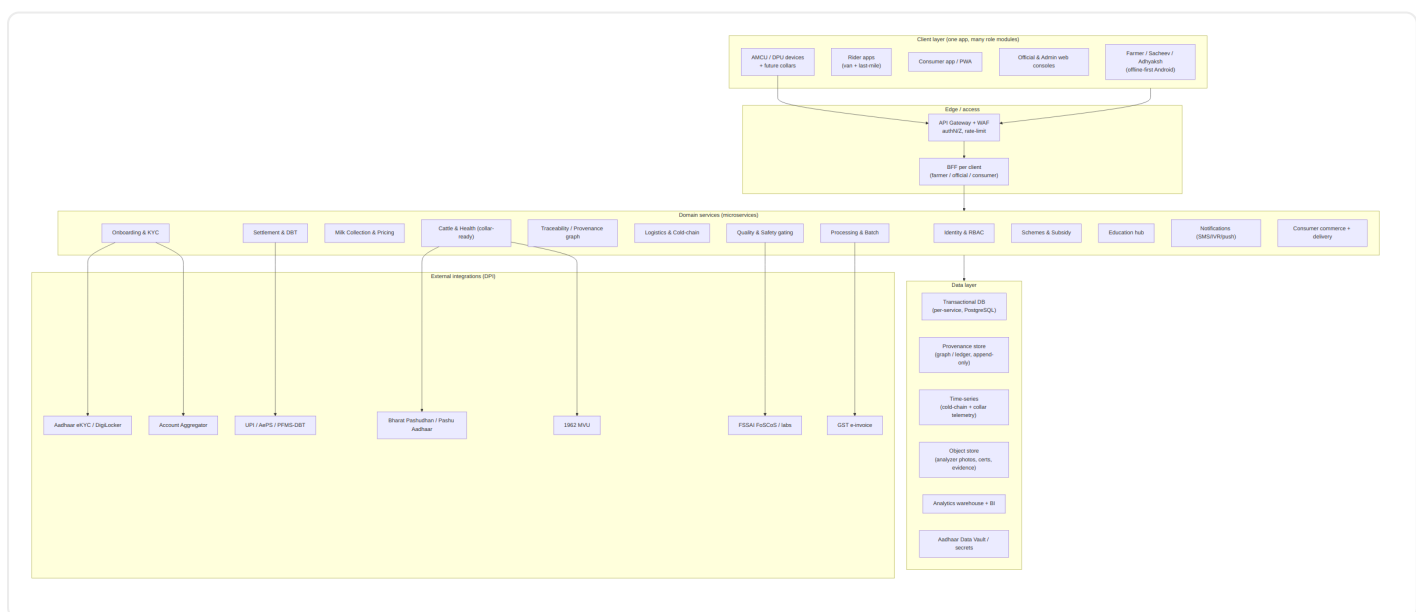
## 5.2 Role catalog

| #  | Role (code)           | Real-world equivalent                           | Tier           | Core permissions                                                                                           | Dashboard focus                                        | KYC tier |
|----|-----------------------|-------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------|
| 1  | FARMER                | Dugdh Utpadak / milk producer                   | Village        | View own pours, bills, payments; own cattle; scheme apply; education; call 1962                            | Simple: today's bill, month total, my cattle, get help | Farm er  |
| 2  | SAMITI_SECRETARY      | DCS Secretary                                   | Village        | Operate AMCU; record pours; run quality tests; generate DCS consignment; society a/c view; trigger payouts | Collection console, can dispatch, member ledger        | High     |
| 3  | SAMITI_ADHYAKSH       | DCS Chairman (elected)                          | Village        | Approve members, oversight, society reports; co-sign governance actions                                    | Society health, member roster, approvals               | High     |
| 4  | MILK_TESTER           | DCS fat/quality tester (sometimes separate)     | Village        | Record analyzer readings / photo-OCR confirm                                                               | Test entry screen                                      | Standard |
| 5  | LRP                   | Local Resource Person (NDDDB village extension) | Village        | Assist onboarding; ration/feed advisory; data entry help                                                   | Field-worker toolkit                                   | Standard |
| 6  | AI_TECH               | AI/Maitri technician (artificial insemination)  | Village        | Log AI/breeding events against Pashu Aadhaar → push to Bharat Pashudhan                                    | AI logbook                                             | Standard |
| 7  | VAN_RIDER             | Collection tanker driver                        | Logistics      | Scan/collect DCS consignments; log pickup time + temp; route manifest                                      | Route + pickups + cold-chain log                       | Rider    |
| 8  | DELIVERY_RIDER        | Last-mile consumer delivery                     | Logistics      | Accept orders, delivery status, proof-of-delivery                                                          | Delivery queue, navigation                             | Rider    |
| 9  | BMC_OPERATOR          | Bulk Milk Cooler / chilling centre operator     | Union          | Pool cans into silo lot; log chilling temp; rapid quality (incl. aflatoxin/coliform strip)                 | Intake + cold-chain + quality                          | Standard |
| 10 | UNION_FIELDSUPERVISOR | Procurement/field officer                       | Union          | Oversee cluster of DCS + BMCs; approve rates; investigate anomalies                                        | Cluster ops, exceptions                                | High     |
| 11 | UNION_PRESIDENT       | Milk Union / Sangh President                    | Union          | District oversight; approvals; reporting                                                                   | Union KPIs, procurement, sales                         | High     |
| 12 | PLANT_OPERATOR        | Processing plant operator                       | Plant          | Receive tankers, create processing batches, link lots, product output                                      | Batch console                                          | High     |
| 13 | PLANT_LAB_ANALYST     | QC lab technician                               | Plant          | Full lab tests (AFM1 ELISA/HPLC, coliform, antibiotics, phosphatase); pass/fail batch; issue QR            | Lab, batch QC certificates                             | High     |
| 14 | PCDF_ADMIN            | Federation administrator                        | State          | State-wide config, rate policy, product master, brand/QR governance                                        | Federation control tower                               | High     |
| 15 | MISSION_OFFICIAL      | Nand Baba Dugdh Mission / Dairy Dept official   | State/District | Scheme + subsidy oversight; DBT status; yield analytics                                                    | Mission dashboard, DBT, yield/heatmaps                 | High     |

| #  | Role (code)       | Real-world equivalent             | Tier     | Core permissions                                                                     | Dashboard focus                  | KYC tier  |
|----|-------------------|-----------------------------------|----------|--------------------------------------------------------------------------------------|----------------------------------|-----------|
| 16 | DISTRICT_VERIFIER | District-level scheme committee   | District | Verify scheme applications + documents                                               | Application queue                | High      |
| 17 | VETERINARIAN      | 1962 MVU vet                      | Health   | Access animal history (consented); e-prescription; disease report → Bharat Pashudhan | Case queue, animal record        | High      |
| 18 | MVU_DRIVER        | MVU ambulance driver/para-vet     | Health   | Dispatch, location, visit log                                                        | Dispatch + route                 | Standard  |
| 19 | CONSUMER          | Retail app user                   | Consumer | Scan QR traceability; buy Parag products; track delivery                             | Product provenance, shop, orders | Minimal   |
| 20 | SUPPORT_AGENT     | Helpdesk                          | Platform | Assist users, raise tickets, limited PII view (audited)                              | Ticketing, user lookup           | High      |
| 21 | STATE_AUDITOR     | Read-only oversight/audit         | Platform | Read-only across scope; export audit logs                                            | Audit + compliance views         | High      |
| 22 | SUPER_ADMIN       | Platform admin                    | Platform | Full config, role grants, feature flags, break-glass (audited, MFA, dual-control)    | Admin console                    | Highest   |
| 23 | SERVICE_ACCOUNT   | Machine (AMCU device, connectors) | System   | Scoped API tokens for device ingest / DPI connectors                                 | n/a (API)                        | mTLS/keys |

**Recommendation on "which profiles to create first":** MVP = roles 1, 2, 3, 7, 12, 13, 19, 22 (the minimum to prove the pour→payment→batch→QR loop). Roles 5, 6, 9, 10, 11, 15, 17 unlock the *mission value* (extension, health, oversight) and come in Phase 2. See roadmap §19.

## 6. High-level system architecture



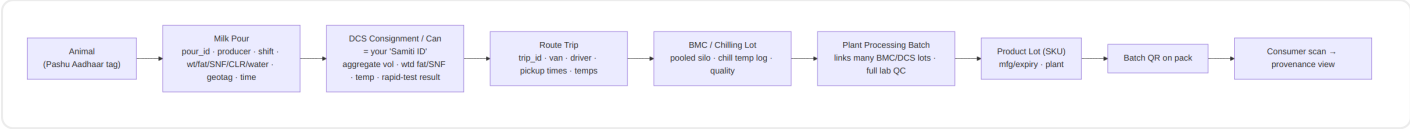
**Architectural stance:** event-driven microservices behind a gateway, **BFFs** tailoring payloads per client (the farmer app payload is radically simpler than the official console), an **append-only**



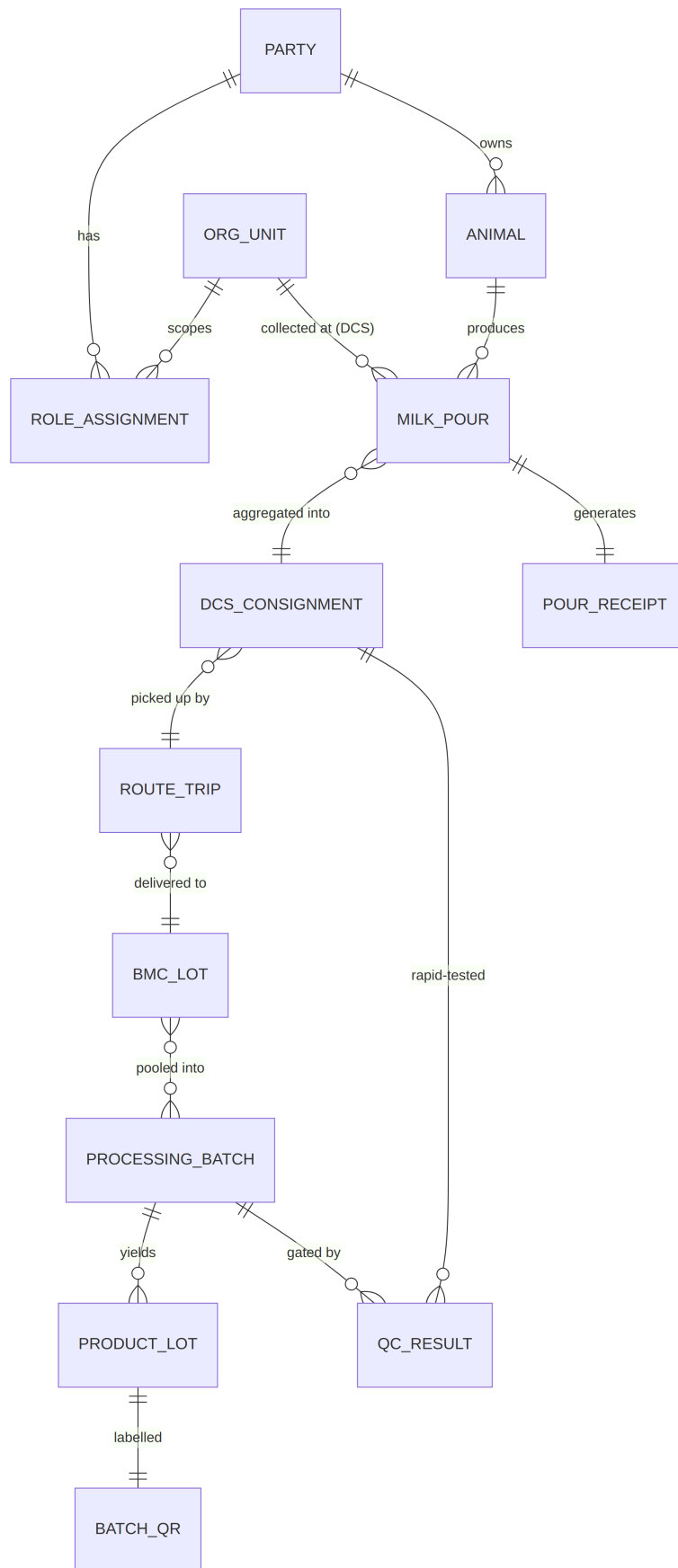
**provenance store** as the source of truth for traceability, and a **thin, well-guarded integration edge** to the DPI systems.

## 7. Milk traceability — the provenance graph

### 7.1 The ID chain



## 7.2 Entity model (simplified ER)



## 7.3 Why append-only / ledger-style

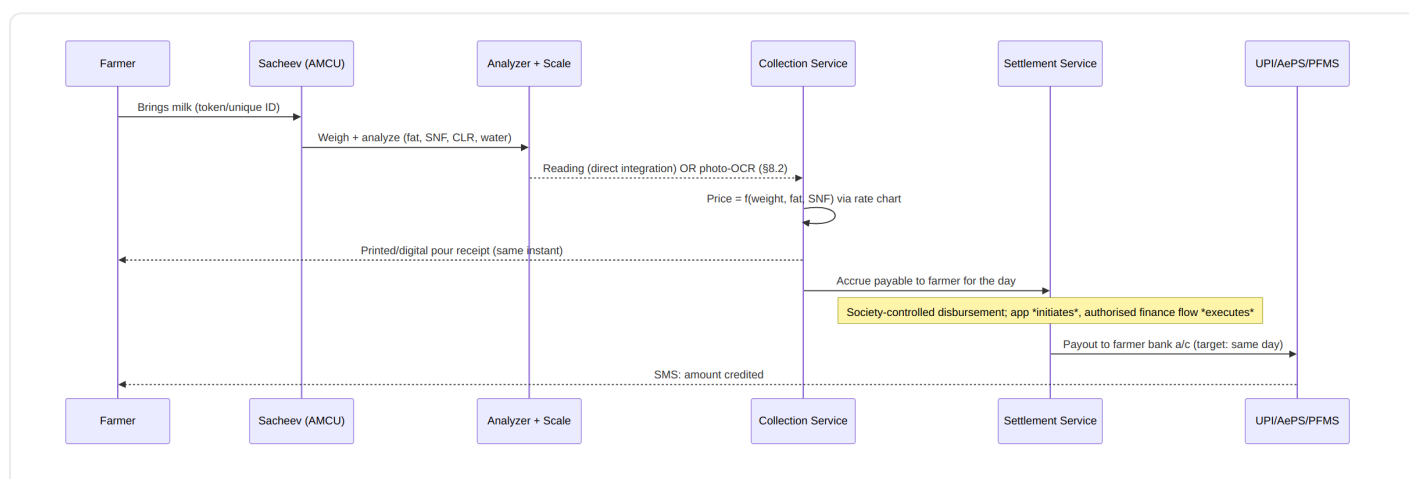
Traceability is only trustworthy if it cannot be quietly rewritten. Store provenance edges as **immutable events**; corrections are *new* events that supersede (never overwrite) prior ones. This gives you audit-grade recall capability (FSSAI food-recall requirement) and tamper-evidence. A graph database (or a hash-chained event log projected into a graph) fits the many-to-many pooling naturally.

## 7.4 The honest limit of pack-level traceability

Because milk is **pooled** at the can, BMC, and plant silo, a finished pack does **not** trace to one farmer. It traces to the **set of contributing samitis** in that batch (with approximate volume proportions). Design the consumer view accordingly: *"This pack's batch was made from milk collected on 3–4 Jul from N samitis in Barabanki district"* + quality certificate. **Samiti-level granularity is achievable and truthful; per-farmer-per-pack is not.** Don't promise the impossible in the UI or to the government.

# 8. Key end-to-end flows

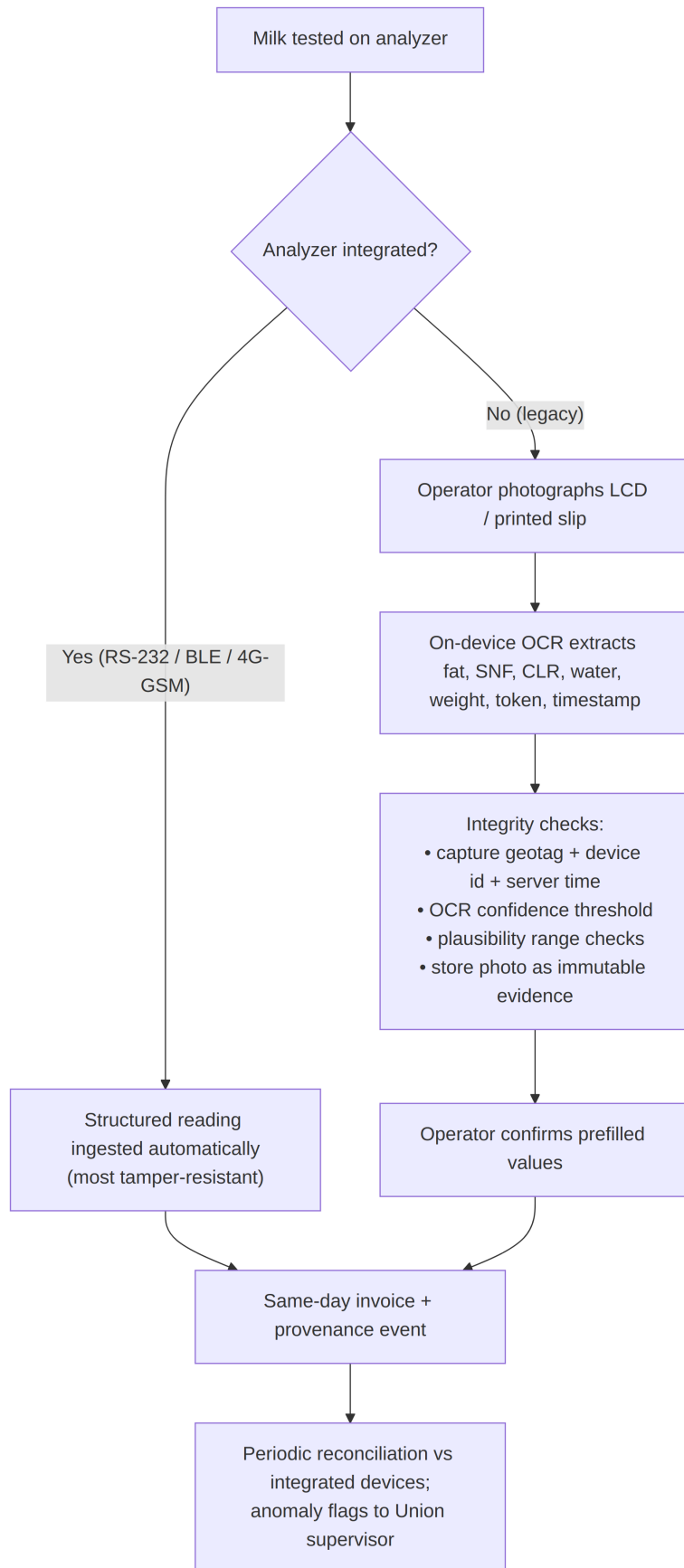
## 8.1 Daily pour → same-day invoice → payment



**Payment guardrail:** Saathi *computes and initiates* settlement and provides the rail integration; it never moves funds autonomously outside the cooperative's authorised financial controls, and it renders subsidy money strictly through **DBT/PFMS**. (Money-movement remains an authorised human/finance action — see §18-B.)

## 8.2 "Photo of the analyzer screen" → autofill (with anti-tamper)

Two ingestion paths — prefer the first, support the second as a bridge for un-integrated legacy analyzers:

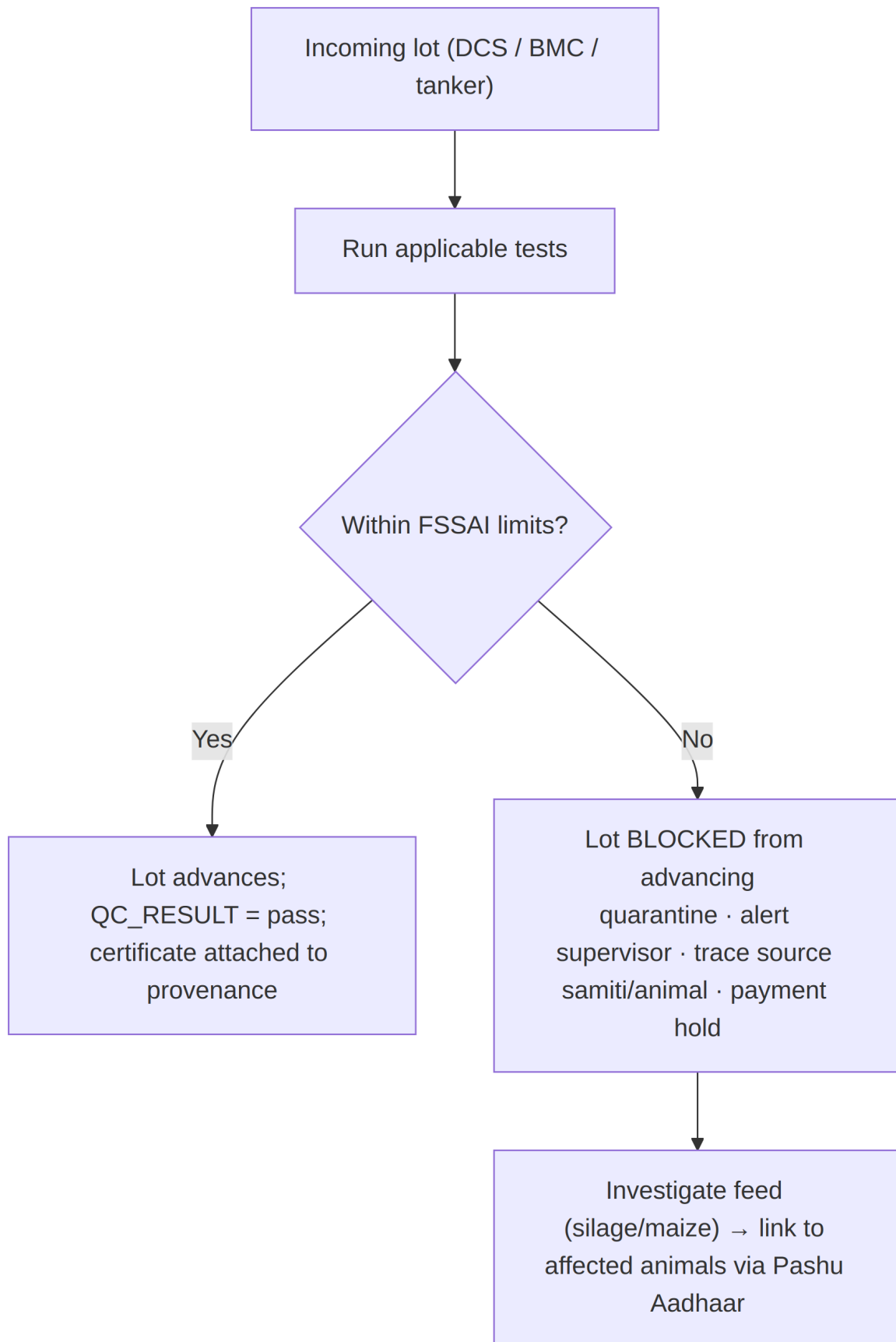


**Be honest with stakeholders:** a photo of a screen *is* spoofable. The controls above (evidence retention, geotag/time binding, plausibility bounds, reconciliation, anomaly alerts) mitigate but don't eliminate fraud. Direct device integration should be the funded end-state; photo-OCR is the on-ramp that gets un-wired samitis onto the platform on day one.

### 8.3 Safety gate — aflatoxin & coliform (the "disease-free milk" enforcement)

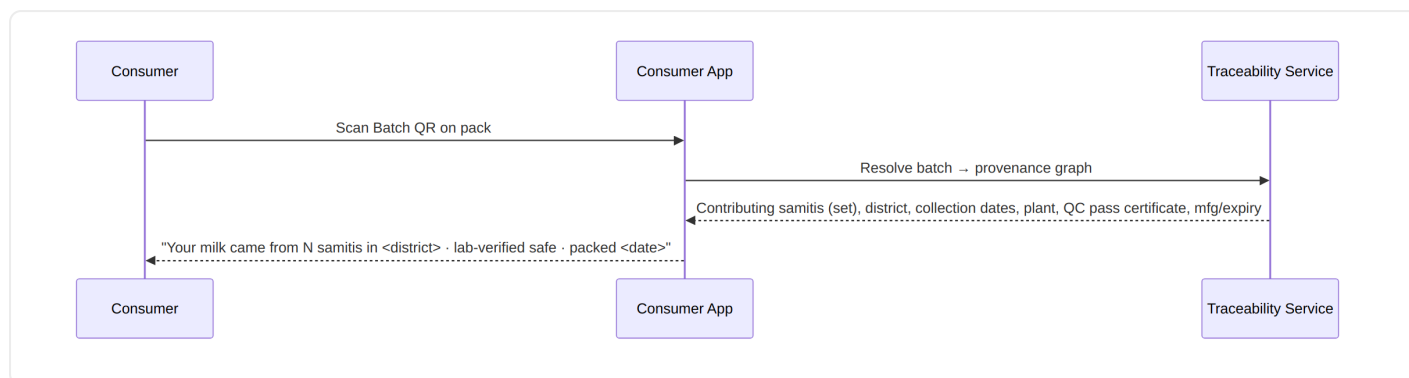
Critical science: **Aflatoxin M1 is heat-stable and survives pasteurisation** — it *must* be caught upstream, not at the finished product. So gate at multiple points against **FSSAI limits**:

| Test                               | FSSAI limit (liquid milk)            | Where tested                                          | Method                      |
|------------------------------------|--------------------------------------|-------------------------------------------------------|-----------------------------|
| Aflatoxin M1                       | $\leq 0.5 \mu\text{g/kg}$            | BMC + Plant (rapid strip at BMC; ELISA/HPLC at plant) | ELISA screen → HPLC confirm |
| Coliform                           | $\leq 10 \text{ CFU/ml}$             | BMC + Plant                                           | Rapid + lab culture         |
| TPC                                | per FSSAI grade                      | Plant                                                 | Lab                         |
| Antibiotic residues                | MRLs (e.g. tetracycline 0.1 mg/kg)   | Plant                                                 | LC-MS/MS                    |
| Phosphatase (pasteurisation check) | must be negative post-pasteurisation | Plant                                                 | Enzyme test                 |



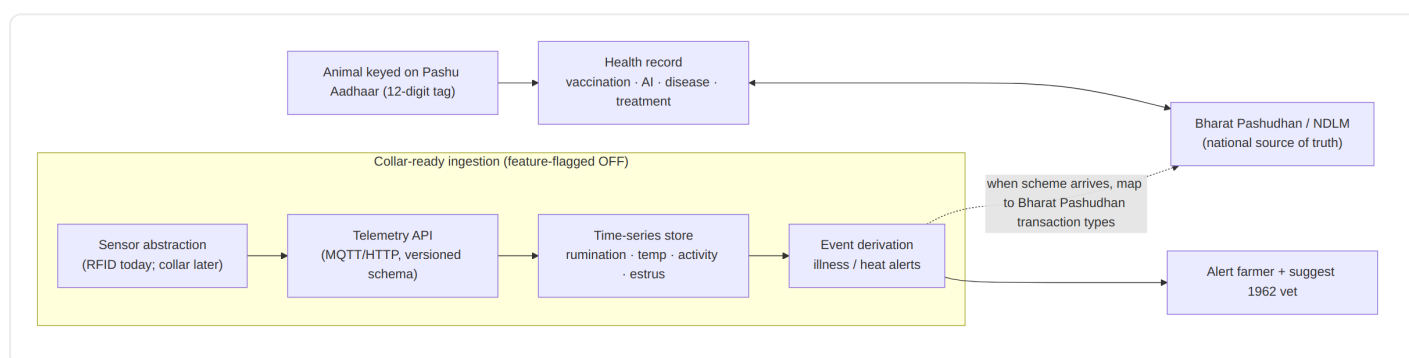
**Because Saathi links lots back to source via the provenance graph, a failed aflatoxin test can be traced to the contributing samiti(s) and — where farm-level testing exists — toward specific feed sources/animals.** That closes the loop between "safety gate" and "root cause".

## 8.4 Consumer scan



## 9. Cattle health & the collar "backdoor"

You correctly noted: **there is no government collar/wearable scheme yet**, so you can't get that data now — but you want a clean hook. Do it via **Bharat Pashudhan / Pashu Aadhaar**, and provision the telemetry path dormant.



**Design specifics for the team:** - Model `ANIMAL.pashu_aadhaar` as the join key across *all* health data so Saathi records are natively interoperable with the national stack. - Ship a **versioned telemetry ingestion contract** (device → MQTT/HTTPS → time-series) and a `collar_enabled` **capability flag** per animal/region — **OFF** now. No schema migration needed when a scheme lands; flip the flag and provision devices. - Today's **RFID** ear-tag readings use the *same* sensor abstraction, so the "backdoor" is already exercised in production with a trivial payload. - Derived health events (illness, estrus) map to Bharat Pashudhan transaction types for push-back to the national DB, and locally trigger a **1962** vet suggestion.

## 10. Education hub, 1962 & veterinary care

- **Education hub:** short vernacular video/audio content (clean milking, feed/ration balancing, breed care), works offline (pre-cached), icon-navigable. Mirror the content model of the national **Pashupalak/1962** apps so material is consistent and re-usable.
- **1962 integration:** in-app "call MVU" button + request tracking (dispatch → arrival → visit log), aligned to the toll-free MVU service. Vet visit outcomes write to the animal's health record (and Bharat Pashudhan).
- **Veterinary care:** e-prescription and disease reporting by `VETERINARIAN`, consented access to animal history, disease-outbreak surveillance signals for `MISSION_OFFICIAL`.

## 11. Consumer app + last-mile delivery

- **Consumer commerce:** browse/buy Parag products; QR provenance (§8.4); order tracking. Minimal PII (mobile OTP), no Aadhaar unless required for a transaction.
- **Last-mile delivery:** `DELIVERY_RIDER` module — order queue, route/navigation, **proof-of-delivery**, cold-chain note for perishables. Reuse the same logistics/cold-chain service as van collection where possible.
- Consider **ONDC** compatibility for the commerce layer (optional, Phase 3) to plug Parag into the open network.

## 12. Admin & governance console

- **SUPER\_ADMIN** : tenant/org config, role grants, **feature flags** (collar, ONDC, per-district rollout), product & rate masters, **break-glass** access — all under **MFA + dual-control + full audit**.
- **MISSION\_OFFICIAL** / **PCDF\_ADMIN** : state control tower — yield-per-cow heatmaps, procurement vs sales, DBT/subsidy status, safety-failure trends, samiti league tables.
- **STATE\_AUDITOR** : read-only + immutable **audit-log export** (who saw/changed what, when) — required for government accountability and DPDP breach forensics.
- **Tailored analytics:** each role sees only its scope (a Sacheev sees their samiti; a Union president their district; the Mission the whole state).

## 13. Integration architecture (external systems)

| System                        | Purpose in Saathi                               | Integration mode    | Notes                                              |
|-------------------------------|-------------------------------------------------|---------------------|----------------------------------------------------|
| Aadhaar eKYC (KUA/AUA)        | Identity for onboarding                         | OTP + Offline XML   | Mask 8 digits; Data Vault; never mandatory-to-deny |
| DigiLocker                    | Land/caste/vehicle docs for scheme + rider KYC  | Requester API       | MeitY approval needed                              |
| Account Aggregator (Sahamati) | Consented financial data (credit/insurance)     | FIU integration     | Sahamati certification                             |
| UPI / AePS                    | Farmer payouts                                  | Via licensed PA     | AePS suits low-connectivity cash-out               |
| PFMS / DBT                    | Subsidy disbursement (Nand Baba schemes)        | Govt DBT rail       | Aadhaar-linked beneficiary a/c                     |
| Bharat Pashudhan / NDLM       | Animal identity + health interoperability       | API (Pashu Aadhaar) | Source of truth for animals                        |
| 1962 MVU                      | Vet dispatch + records                          | API / deep-link     | Align with national service                        |
| FSSAI FoSCoS + NABL labs      | Licensing + confirmatory test results           | API / manual upload | Recall + certification                             |
| GST e-invoice (IRP)           | B2B (DCS→Union) invoices, product GST           | IRP API             | e-invoicing thresholds                             |
| SMS / IVR gateway             | Vernacular notifications for low-literacy users | Provider API        | IVR/voice is essential, not optional               |



## 14. Data architecture

| Store                        | Technology (recommendation)                             | Holds                                    |
|------------------------------|---------------------------------------------------------|------------------------------------------|
| Transactional (per service)  | <b>PostgreSQL</b> (managed, in-India region)            | Parties, roles, pours, batches, orders   |
| Provenance / traceability    | <b>Graph DB or hash-chained event log</b> (append-only) | The pour→QR graph, immutable             |
| Time-series                  | <b>Time-series DB</b> (e.g. Timescale/Influx)           | Cold-chain temps, collar/RFID telemetry  |
| Object store                 | <b>S3-compatible, in-India</b>                          | Analyzer photos, QC certs, PoD, evidence |
| Analytics warehouse          | <b>Columnar DWH + BI</b>                                | Yield, procurement, safety trends, DBT   |
| Secrets / Aadhaar Data Vault | <b>HSM-backed vault</b>                                 | Aadhaar refs (tokenised), keys           |
| Offline (device)             | <b>Local encrypted DB + sync engine</b>                 | Collection-point offline queue           |

**Data residency:** all stores in **India** (MeitY-empanelled CSP / GI-Cloud / NIC). See §18-D.

## 15. Recommended tech stack

- **Mobile (farmer/rider/Sacheev):** Android-first, **offline-first** with a robust local DB + background sync; **Kotlin** (or Flutter for shared UI). PWA for consumer to lower install friction.
- **Web consoles (officials/admin):** React/TypeScript; role-scoped module federation.
- **Backend:** microservices in a memory-safe/typed stack (**Go / Java-Kotlin / TypeScript-Node**), event bus (**Kafka/NATS**), gateway + WAF.
- **Provenance:** append-only event store projected to a graph (Neo4j/JanusGraph) or ledger-style with hash chaining.
- **ML/OCR (analyzer photos):** on-device OCR (ML Kit/Tesseract-class) for offline autofill; server-side re-verification.
- **Infra:** containerised (Kubernetes) on a **MeitY-empanelled Indian cloud** or NIC/GI-Cloud; IaC; observability (metrics/logs/traces); NTP-synced (CERT-In).

Treat this as a *reference* stack — the non-negotiables are: offline-first clients, append-only provenance, in-India hosting, DPI integration, and audit-grade logging. The specific languages/DBs are swappable.

## 16. Non-functional requirements

| NFR                               | Target / approach                                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Offline-first</b>              | Every collection-point + rider action works fully offline; conflict-free sync; queued payouts reconcile on reconnect                               |
| <b>Low-literacy accessibility</b> | Icon-first, <b>voice prompts + IVR</b> , full vernacular (Hindi + regional), large touch targets, minimal text; <b>WCAG 2.1 AA</b> (GIGW-mandated) |
| <b>Scale</b>                      | Design for millions of farmers, ~18,000+ DCS, twice-daily pours → peak write bursts at shift times; horizontal scaling, idempotent ingest          |
| <b>Performance</b>                | Sub-second pour receipt; batching for sync; edge caching for education content                                                                     |
| <b>Reliability / DR</b>           | Multi-AZ in-India; defined <b>RPO/RTO</b> ; business continuity plan; provenance store never loses events                                          |
| <b>Tamper-evidence</b>            | Append-only + hash chaining + evidence retention (\$7.3, \$8.2)                                                                                    |
| <b>Observability &amp; audit</b>  | Full audit trail (who/what/when), immutable, exportable                                                                                            |
| <b>Localization</b>               | Unicode; MDDS metadata standards; number/date formats                                                                                              |

## 17. Security architecture (summary)

- **AuthN**: phone-OTP + device binding for field users; Video-KYC-backed accounts for officials; **MFA** for admin.
- **AuthZ**: org-scoped RBAC (+ ABAC for fine cases); least privilege; role validity windows; break-glass with dual-control.
- **Data protection**: encryption in transit (TLS) and at rest; **Aadhaar Data Vault** + first-8-digit masking; PII tokenisation; field-level access controls.
- **Device/API security**: mTLS + scoped tokens for AMCU/service accounts; **OWASP MASVS** (mobile) + **ASVS/Top-10** (services); rate-limiting + WAF at edge.
- **Logging**: in-India log retention **≥180 days** (CERT-In); NTP sync; tamper-evident audit log.
- **Incident response**: **6-hour** CERT-In breach reporting; DPDP breach notification to Data Protection Board + affected principals; UIDAI notification if Aadhaar data touched.
- **Assurance**: pre-go-live **VAPT by CERT-In-empanelled auditor** + **STQC** security certification; annual audits.

## 18. Compliance register (industry-grade, government-app)

This is the "all the compliances" list. Grouped by domain; treat each as a work item with an owner. A platform of this scale will very likely be classified a **Significant Data Fiduciary (SDF)** under DPDP — plan for the heavier obligations from day one.

### A. Data protection & privacy

- **DPDP Act 2023 + DPDP Rules 2025** — multilingual plain-language **consent + notice**; purpose limitation; **data minimisation**; data-principal rights (access, correction, **erasure**, grievance) with SLA; **Consent Manager** integration; **breach notification** to Data Protection Board + affected users "without delay"; appoint **DPO** + grievance officer.

- **SDF obligations (assume applicable):** annual **DPIA + independent audit**; algorithmic-fairness/risk review of any automated decisioning; **data-localisation** for any data categories the government notifies; report significant DPIA/audit observations to the Board.
- **Children's data (under-18):** verifiable parental consent; **no behavioural tracking / targeted ads** to minors (relevant if any minor users or minor-linked records).
- **Aadhaar Act + UIDAI regulations:** mask first 8 digits; **Aadhaar Data Vault** for stored references; no biometric storage unless authorised AUA/KUA; local-language purpose notice; **Virtual ID** support; **never deny service for lack of Aadhaar**; breach → UIDAI.
- **IT Act 2000** (Sec 43A, 66, 72A; Sec 70 if declared a "**Protected System**") and transitional IT (SPDI) Rules 2011 until DPDP fully supersedes.
- **CERT-In Directions (2022):** **6-hour** incident reporting; **180-day** in-India log retention; NTP sync; KYC/records maintenance.

## B. Financial & payments

- **RBI** — route all money-movement through a **licensed Payment Aggregator**; nodal/escrow accounts; card tokenisation (no full-PAN storage); **RBI KYC Master Direction** for account linking/penny-drop; **PMLA** where applicable.
- **NPCI** — UPI/AePS/IMPS circulars for payouts.
- **PFMS / DBT** — Aadhaar-linked beneficiary accounts for **Nand Baba** subsidy disbursement.
- **Account Aggregator (Sahamati)** framework if pulling consented financial data.
- **GST** — e-invoicing (IRP) for B2B (DCS→Union) and product GST; TDS/TCS as applicable.

## C. Food safety & dairy standards

- **FSSAI** — FSS Act 2006; **FSS (Food Products Standards) Regs 2011; Licensing & Registration** (plant/BMC on **FoSCoS**); enforce **AFM1 ≤0.5 µg/kg**, **coliform ≤10 CFU/ml**, TPC, **antibiotic MRLs**, **phosphatase** (pasteurisation), **Vit A/D fortification**; **food-recall plan** + traceability.
- **BIS / IS standards** — IS 1166 (milk), IS 3508 (ghee), IS 13690 (butter), IS 10484 (paneer); **AGMARK** where used; **Codex Alimentarius** alignment for exports.
- **Legal Metrology Act 2009 + Packaged Commodities Rules** — **weighing scales & milk analyzers must be verified/stamped** (legally required); net-quantity, MRP, and unit-sale declarations on packs.

## D. Government IT / e-Governance

- **GIGW 3.0** (Guidelines for Indian Government Websites & Apps) — quality, security, accessibility conformance.
- **WCAG 2.1 AA** accessibility (GIGW-mandated) — essential for low-literacy/differently-abled farmers.
- **STQC certification + VAPT by CERT-In-empanelled auditor** before go-live; periodic re-audit.
- **Hosting: MeitY-empanelled Cloud Service Provider / GI-Cloud (MeghRaj) / NIC; data residency in India.**
- **IndEA / India Enterprise Architecture + Open API policy (GoI) + NDSAP** for data sharing; **MDDS** metadata standards; Unicode/Indian-language support; optional **UMANG** listing.
- **ISO 27001** (ISMS), **ISO 27701** (privacy), **SOC 2** for vendors; **e-Sign / DSC** (CDAC) for signed official records.

## E. Sector / operational

- **Cooperative Societies Act (UP)** — DCS/Union/Federation governance, **election records**, statutory audit.
- **National Digital Livestock Mission (NDLM)** data standards — **Pashu Aadhaar** interoperability with Bharat Pashudhan.
- **Prevention of Cruelty to Animals Act** — welfare data handling.
- **Code on Social Security 2020** (gig/platform-worker provisions) for van + delivery riders; **IRDAI** if brokering cattle insurance.

## F. Engineering & quality

- **OWASP MASVS** (mobile), **OWASP ASVS / Top-10** (services), secure SDLC.
- **ISO 25010** software quality; **ISO 9001** process; defined **RPO/RTO** + BCP/DR.
- Tamper-evident **audit logging**; append-only provenance; performance & accessibility budgets.

## 19. Phased delivery roadmap

| Phase                              | Goal                            | Roles                 | Key deliverables                                                                                                            |
|------------------------------------|---------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>P0 — Foundation</b>             | Identity + one loop             | 1,2,3,22              | Party/RBAC + onboarding/KYC; role switcher; offline collection console; pour receipt                                        |
| <b>P1 — Traceability + payment</b> | Pour→QR + same-day pay          | +7,12,13,19           | Provenance graph; analyzer direct-integration + photo-OCR bridge; safety gate; batch QR; consumer scan; settlement/DBT rail |
| <b>P2 — Mission value</b>          | Health + oversight + subsidy    | +5,6,9,10,11,15,16,17 | Cattle/health + Bharat Pashudhan sync; collar-ready dormant path; 1962; education hub; scheme/subsidy; official dashboards  |
| <b>P3 — Consumer + scale</b>       | Commerce + delivery + hardening | +8,20,21              | Consumer commerce; last-mile delivery; ONDC (optional); SDF audit; full compliance certification; scale-out                 |

## 20. Open decisions the team needs to resolve

1. **Build-vs-federate with NDDDB AMCS / i-DIS** where already deployed — ingest their data or run Saathi's own collection console per DCS? (Affects §6, §13.)
2. **Provenance store choice** — native graph DB vs hash-chained event log projected to graph (both meet append-only; different ops profiles). (§7.3)
3. **Payment settlement authority** — exact approval flow for who signs off same-day disbursement, and PA vs direct cooperative banking rail. (§8.1, §18-B)
4. **SDF classification** — confirm with legal; if confirmed, DPIA + annual audit + localisation are mandatory from launch. (§18-A)
5. **Analyzer fleet reality** — what % of target samitis have integrable (RS-232/BLE/GSM) analyzers vs legacy-only? Determines how heavily the photo-OCR bridge is loaded. (§8.2)
6. **Traceability granularity promise** — align UI + government MoU on **samiti-level** (achievable) vs any per-farmer pack claim (not achievable post-pooling). (§7.4)
7. **Weights & Measures** — process for periodic Legal-Metrology verification/stamping of scales & analyzers feeding the system. (§18-C)